NATIONAL ECONOMICS UNIVERSITY
SCHOOL OF TECHNOLOGY

FINAL PROJECT REPORT

Course: Data Analysis with Python

PROJECT NO. 02
TELCO CUSTOMER CHURN PREDICTION

Class: DS6A
Authors: Group 3

No.	Full Name	Student ID
1	Đinh Vân Anh	11247123
2	Phạm Thùy Anh	11247137
3	Nguyễn Hoàng Hà	11247162
4	Hoàng Thị Lệ Xuân	11247249

Project Repository: GitHub Link
Dataset Source: Kaggle Link

Instructor: Dr. Tran Hung
Hanoi, May 2026

Contents

List of Figures

List of Tables

1 Introduction

1.1 Problem Statement

Customer churn represents one of the most critical challenges in the telecommunications industry. It occurs when subscribers discontinue their service with a provider. High churn rates directly lead to revenue loss and increased customer acquisition costs, which are significantly higher than retention costs.

1.2 Project Objectives

The goal of this project is to analyze the factors influencing customer behavior and build a predictive model to identify customers at high risk of leaving the service. By doing so, the company can implement targeted retention strategies.

1.3 Problem Type

This is a binary classification problem where the objective is to predict the `Churn` status:

- `Yes`: The customer discontinued the service.
- `No`: The customer remained with the service.

2 Understand the problem and the dataset

2.1 Download dataset from Kaggle

The dataset is downloaded directly into the working environment using the Kaggle API (`kaggle datasets download`). This approach ensures data reproducibility. The downloaded archive is unzipped to extract the raw data file: `WA_Fn-UseC_-Telco-Customer-Churn.csv`.

2.2 Import libraries and Load the data

Essential Python libraries for data manipulation and visualization are imported, including `pandas`, `numpy`, and `matplotlib.pyplot`. The dataset is loaded into a pandas DataFrame. An initial inspection confirms that the dataset is loaded successfully with a shape of 7,043 rows and 21 columns.

2.3 Problem description and practical meaning

What is customer churn?
Customer churn refers to the phenomenon where customers stop using a company's service. In the context of a telecommunications (telco) company, churn means a subscriber cancels or does not renew their contract.
Why is this problem important in practice?

- Acquiring a new customer is significantly more expensive than retaining an existing one (industry estimates range from 5x to 25x more costly).
- High churn rates directly reduce revenue and signal customer dissatisfaction.
- By predicting which customers are likely to churn, the company can proactively offer targeted retention incentives (discounts, improved plans, personalized support) before the customer leaves.

Problem type: Binary classification — predicting whether a customer will churn (`Yes`) or not (`No`).

2.4 Target variable identification

The target variable for this analysis is `Churn`. An inspection of its value counts reveals the following distribution:

- **No**: 5,174 customers (73.46%)
- **Yes**: 1,869 customers (26.54%)

Interpretation: 'Yes' means the customer churned (left the company), while 'No' means the customer is still active. The dataset is moderately imbalanced, with churned customers representing approximately 26% of the total. This imbalance must be considered during the modeling phase.

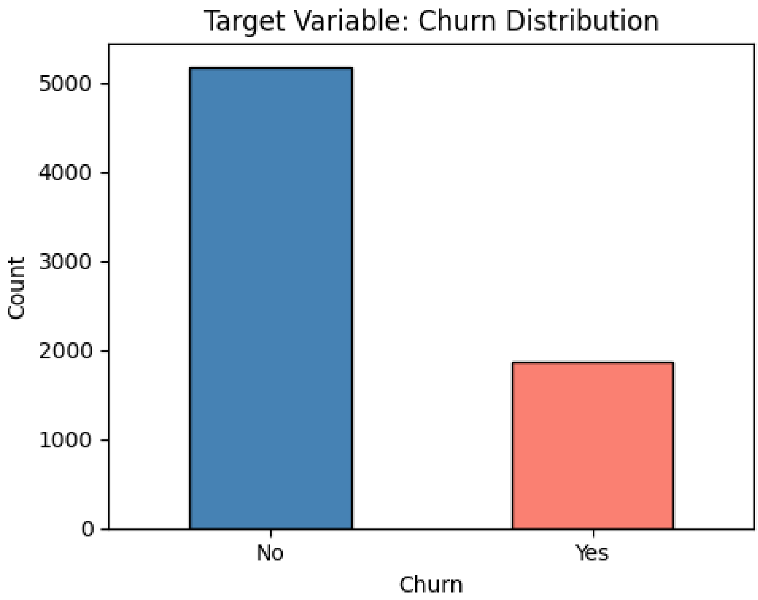

Figure 1: Target Variable: Churn Distribution

2.5 Feature Overview

The dataset consists of 21 columns which can be logically grouped into four main categories. The table below provides a structured overview of these features and their business relevance.

Category	Features	Description
Demographics	`gender`, `SeniorCitizen`, `Partner`, `Dependents`	Basic personal information about the customer.
Account Info	`tenure`, `Contract`, `PaperlessBilling`, `PaymentMethod`, `MonthlyCharges`, `TotalCharges`	Details regarding the customer's billing and loyalty.
Phone Services	`PhoneService`, `MultipleLines`	Subscription status for phone-related services.
Internet Services	`InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`	Subscription status for internet and additional value-added services.
Target Variable	`Churn`	Indicates whether the customer left within the last month.

Table 1: Logical categorization of dataset features.

Specifically, the following key features were identified as primary predictors during initial inspection:

- **tenure:** Duration of the customer's relationship with the company (in months).

- **MonthlyCharges:** The amount charged to the customer monthly.
- **TotalCharges:** The total amount charged over the customer's lifetime.
- **Contract:** The contract term (Month-to-month, One year, Two year).
- **InternetService:** Type of internet provider (DSL, Fiber optic, No).
- **PaymentMethod:** Method of payment (Electronic check, Mailed check, Bank transfer, Credit card).

Key input features include `tenure` (months with the company), `MonthlyCharges`, `TotalCharges`, `Contract` type, `InternetService` type, and `PaymentMethod`.

2.6 Dataset shape, column names, and data types

The dataset consists of 7,043 rows and 21 columns. A summary of the data types reveals:

- **object (18 columns):** Categorical features, string values, and identifiers.
- **int64 (2 columns):** Integer variables (`SeniorCitizen`, `tenure`).
- **float64 (1 column):** Decimal numerical variables (`MonthlyCharges`).

2.7 First statistical inspection

An initial check for missing values using `isna().sum()` detected 0 standard `NaN` values. Full descriptive statistics were generated for both numeric and categorical columns.
Unique value checks highlighted a known quirk in this dataset: the `TotalCharges` column is stored as an `object` (string) instead of a float. Attempting to convert it to a numeric format revealed 11 rows where `TotalCharges` is a blank space (`" "`).
Furthermore:

- There are 0 fully duplicated rows and 0 duplicate `customerID`s.
- The `SeniorCitizen` column is stored as an integer (`0` = No, `1` = Yes), unlike other binary columns which use string representations.

2.8 Summary of Key Observations

To conclude the initial data understanding phase, the key observations are:

1. **Dataset size:** 7,043 rows × 21 columns.
2. **Target variable:** `Churn` is binary (Yes / No) with a 26% churn rate.
3. **Hidden missing values:** While no standard `NaN` values are visible, `TotalCharges` contains hidden missing values (whitespace strings) and must be converted to `float64`.
4. **Categorical anomalies:** `SeniorCitizen` is stored as an integer and should be treated as a categorical variable.
5. **Uniqueness:** There are no duplicate rows, and `customerID` is a unique identifier with no predictive value.

3 Data Cleaning

3.1 Make a working copy

We never modify the raw dataset directly. All cleaning processes are performed on a working copy (`df = df_raw.copy()`) so we can always compare our results against the original data. The initial shape of the working copy is 7,043 rows and 21 columns.

3.2 Standardize missing-value representations

In this dataset, missing values are not stored as standard `NaN`. Instead, they appear as whitespace strings (e.g., `' '`), particularly in the `TotalCharges` column. We used regular expressions to replace whitespace-only strings and common placeholders (like "none", "N/A", "unknown") with `NaN` so pandas can handle them correctly. After standardization, 11 missing cells were identified in `TotalCharges`.

3.3 Handle missing values

Since `TotalCharges` is a numeric feature, we first converted it using `pd.to_numeric`. Inspection revealed that all 11 rows with missing `TotalCharges` correspond to customers with `tenure = 0` (brand-new customers who have not yet been charged). We imputed these missing values using the median of the existing `TotalCharges` values (1397.47), which is a safer approximation than the mean in the presence of outliers. A final check confirmed that zero missing values remain.

3.4 Remove duplicate rows

The `customerID` serves as the unique identifier for each customer. Duplicate `customerID` values would distort counts and model training. The inspection confirmed that the dataset is completely free of duplicate rows.

3.5 Correct data types

Based on initial inspections, two columns required data type correction:

- `TotalCharges`: Successfully converted from `object` to `float64`.
- `SeniorCitizen`: Initially stored as `int64` (0 and 1) but represents a binary categorical variable. It was converted to string values ('No' and 'Yes') to ensure consistency with other categorical features like `Partner` and `Dependents`.

3.6 Detect and repair invalid / abnormal values

Business rules were defined for numeric columns based on domain knowledge: `tenure` (0 – 72 months), `MonthlyCharges` ($0 – $200), and `TotalCharges` ($0 – $15,000). All data points fell perfectly within these valid ranges. Furthermore, an IQR-based outlier detection and boxplot visualizations showed no anomalous outliers. The data distributions align with logical service behaviors, meaning no values needed to be removed or corrected.

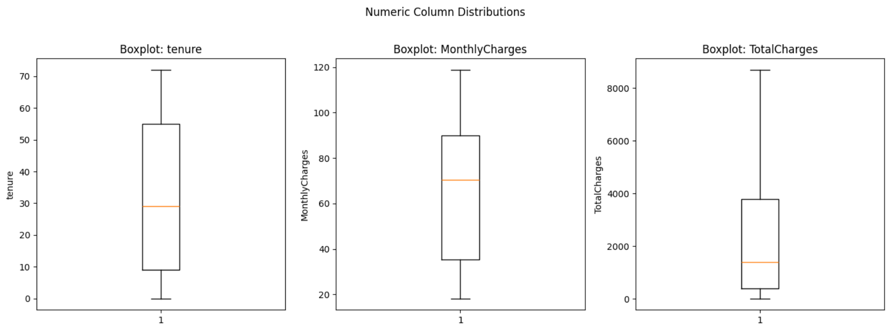

Figure 2: Numeric Column Distributions (Boxplots)

3.7 Unnecessary columns

The `customerID` column is a unique identifier with absolutely no predictive value for modeling. While it is kept in the cleaned dataset for traceability, it is flagged here to be dropped before feeding the data into a machine learning model.

3.8 Standardize categorical text values

We checked all `object`-type columns for inconsistent labels. Leading and trailing whitespaces were stripped from all text columns. An inspection of unique values confirmed that all binary and service-related categorical labels (e.g., 'Yes', 'No', 'No internet service') are already clean and consistent. No additional remapping was required.

3.9 Rule-based validation summary

Before finalizing the cleaning step, the dataset was evaluated against a strict set of business rules, including

4 Exploratory Data Analysis (EDA)

This section provides an in-depth analysis of the dataset through descriptive statistics and data visualization to uncover patterns, anomalies, and relationships between features and the target variable (`Churn`).

4.1 Descriptive Statistics

To gain an initial understanding of the data distribution and central tendencies, we perform a statistical summary for both numerical and categorical features.

4.1.1 Numerical Features Statistics

The table below summarizes the distribution of the primary numerical variables: `tenure`, `MonthlyCharges`, and `TotalCharges`.

Statistic	tenure	MonthlyCharges	TotalCharges
Count	7043.00	7043.00	7043.00
Mean	32.37	64.76	2281.92
Std Dev	24.56	30.09	2265.27
Min	0.00	18.25	18.80
25% (Q1)	9.00	35.50	402.23
50% (Median)	29.00	70.35	1397.48
75% (Q3)	55.00	89.85	3786.60
Max	72.00	118.75	8684.80

Table 2: Descriptive Statistics for Numerical Features.

Key Observations:

- **Tenure:** Average customer loyalty is 32.4 months. However, the high standard deviation (24.6) and the gap between Q1 (9 months) and Q3 (55 months) indicate a polarized customer base consisting of many new users and highly loyal long-term users.
- **Monthly Charges:** The average bill is $64.76. The wide range and interquartile range ($35.50 to $89.85) confirm that customers are spread across various service tiers.
- **Total Charges:** The distribution is heavily right-skewed, as the mean ($2,281.92) significantly exceeds the median ($1,397.48). This suggests that a subset of long-tenure, high-spending customers contributes disproportionately to total revenue.

4.1.2 Categorical Features Statistics

For categorical variables, we analyze the frequency of the dominant values and their respective proportions to identify the majority customer profiles.

Feature	Unique	Top Value	Count	Percentage
gender	2	Male	3555	50.48%
SeniorCitizen	2	No	5901	83.79%
Partner	2	No	3641	51.70%
Dependents	2	No	4933	70.04%
PhoneService	2	Yes	6361	90.32%
MultipleLines	3	No	3390	48.13%
InternetService	3	Fiber optic	3096	43.96%
OnlineSecurity	3	No	3498	49.67%
OnlineBackup	3	No	3088	43.84%
DeviceProtection	3	No	3095	43.94%
TechSupport	3	No	3473	49.31%
StreamingTV	3	No	2810	39.90%
StreamingMovies	3	No	2785	39.54%
Contract	3	Month-to-month	3875	55.02%
PaperlessBilling	2	Yes	4171	59.22%
PaymentMethod	4	Electronic check	2365	33.58%
Churn	2	No	5174	73.46%

Table 3: Descriptive Statistics for Categorical Features.

Profile Insights:

- **Demographics:** The sample is gender-balanced, but predominantly composed of non-senior citizens (83.79%) and individuals without dependents (70.04%).
- **Risk Factors:** More than half of the customers (55.02%) are on *Month-to-month* contracts, which traditionally carries a higher risk of churn. Additionally, *Electronic check* is the most frequent payment method (33.58%), often associated with higher attrition rates.
- **Service Gaps:** While *Fiber optic* is popular, there is low adoption of protective services like *Online Security* or *Tech Support*, potentially impacting long-term retention.

4.2 Chart 1: Distribution of the Target Variable (Churn)

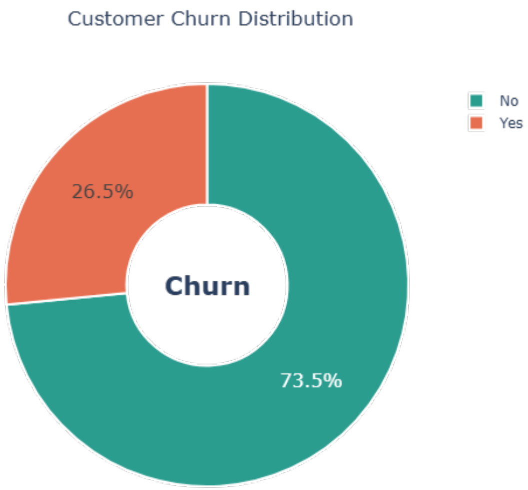

Figure 3: Customer Churn Distribution

Observations: The chart reveals that 26.54% of the customer base has churned, while the remaining 73.46% continue to use the service. This distribution indicates a class imbalance in our target variable. To prevent the predictive model from becoming biased towards the majority class (`No`), this imbalance must be addressed during preprocessing. Despite the imbalance, a churn rate where more than 1 in 4 customers leave is significantly high, signaling a critical customer retention problem.

4.3 Chart 2: Distribution of Numerical Features by Churn

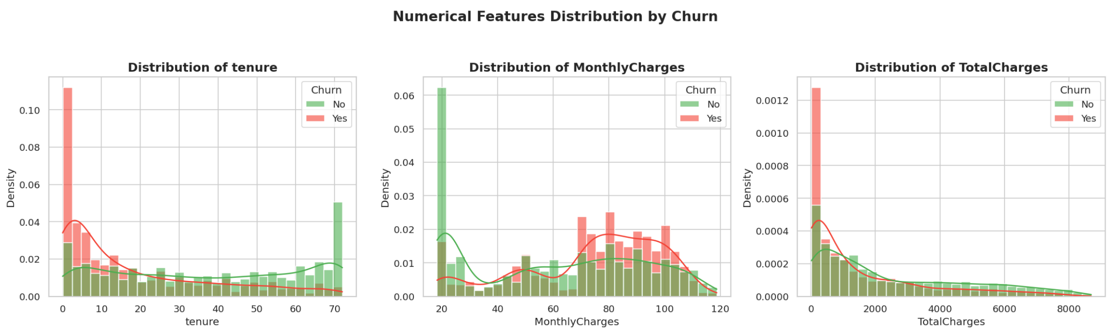

Figure 4: Numerical Features Distribution by Churn

Observations:

- **Tenure:** Churned customers are heavily concentrated at low tenure values (0–10 months). Retained customers are spread more evenly across all tenure ranges. New customers are at the highest risk of churning.
- **Monthly Charges:** Customers who churned tend to have higher monthly charges, skewed toward the \$70–\$100+ range, indicating dissatisfaction with pricing relative to perceived value.
- **Total Charges:** Retained customers show a wider and higher distribution of total charges, consistent with their longer tenure.

4.4 Chart 3: Churn Rate by Key Categorical Features

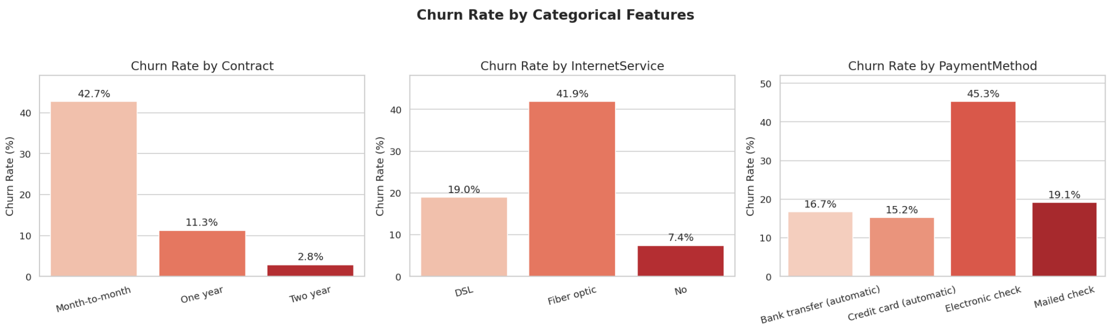

Figure 5: Churn Rate by Categorical Features

Observations:

- **Contract Type:** Month-to-month customers have a churn rate of 42.7%, compared to 11.3% for one-year and 2.8% for two-year contracts.
- **Internet Service:** Fiber optic users exhibit a notably higher churn rate (41.9%) compared to DSL users (19.0%).
- **Payment Method:** Customers paying via electronic check churn at the highest rate (45.3%), while those using automatic payment methods churn significantly less (15–17%).

4.5 Chart 4: Tenure by Churn

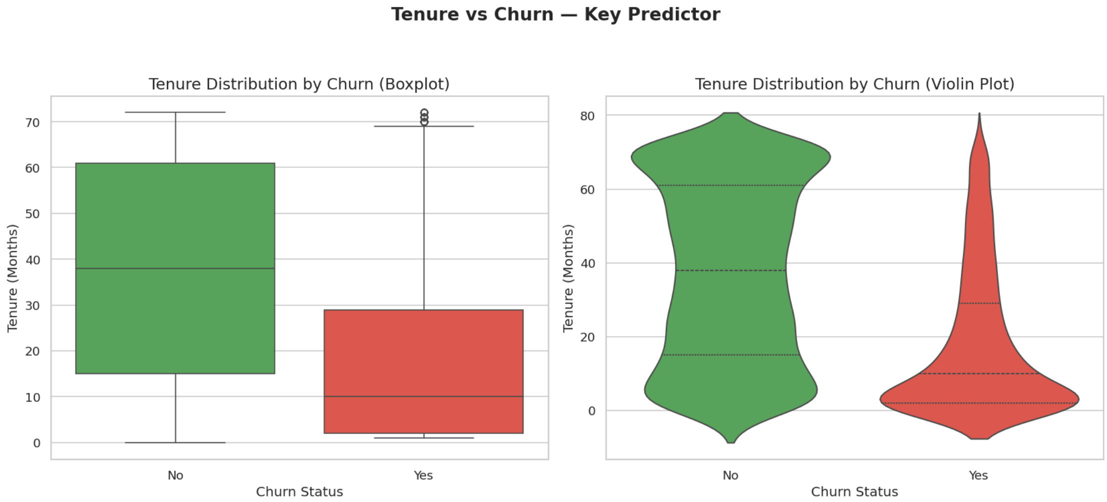

Figure 6: Tenure vs Churn - Key Predictor (Box & Violin Plot)

Observations: The boxplot clearly shows that churned customers have a median tenure of approximately 10 months, compared to 38 months for retained customers. The violin plot reveals that the churn group is heavily concentrated at very low tenure values, tapering off sharply after 20 months. This confirms that the first 12 months are the most critical retention window.

4.6 Chart 5: Correlation Heatmap

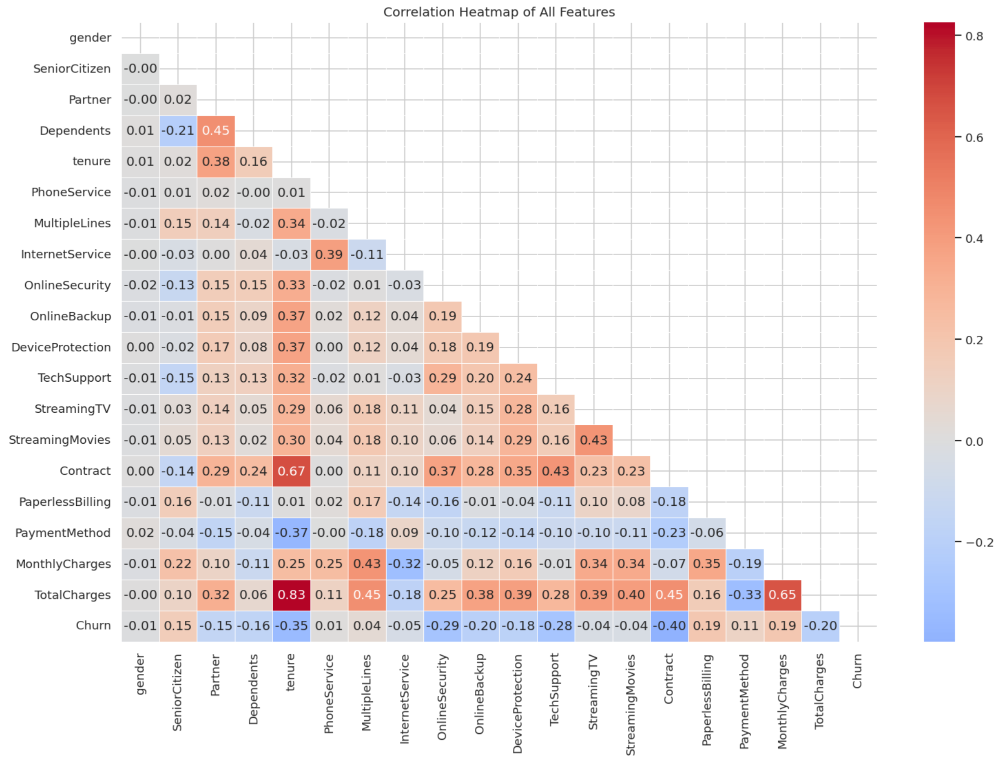

Figure 7: Correlation Heatmap of All Features

Observations:

- **Tenure** shows a strong negative correlation with Churn.
- **Monthly Charges** has a positive correlation with Churn.
- **Total Charges** is highly correlated with tenure ($r \approx 0.83$).
- Service-related features are moderately correlated with each other.
- **SeniorCitizen** shows a mild positive correlation with churn.

4.7 Chart 6: Demographic Factors vs Churn

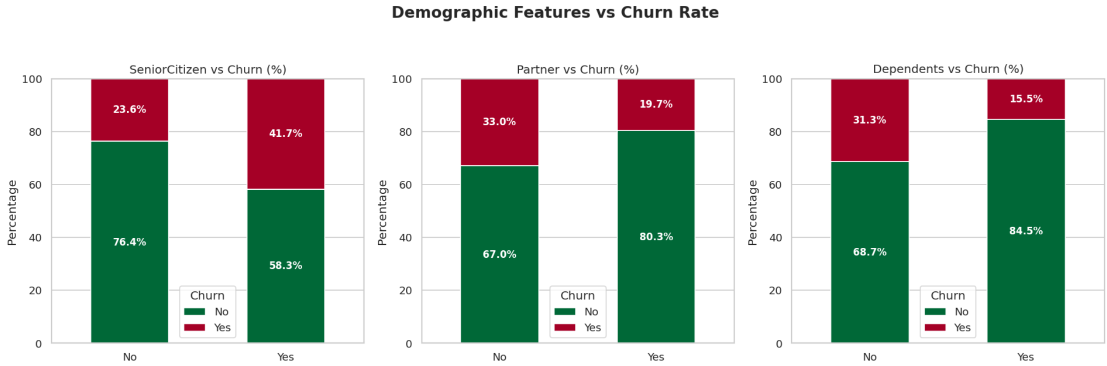

Figure 8: Demographic Features vs Churn Rate

Observations: Senior Citizens have a churn rate of approximately 42%, nearly double the rate of non-senior customers (24%). Customers without a partner or dependents churn at a higher rate than those with them, reflecting lower "stickiness" or household service commitments.

4.8 Chart 7: MonthlyCharges vs Tenure colored by Churn

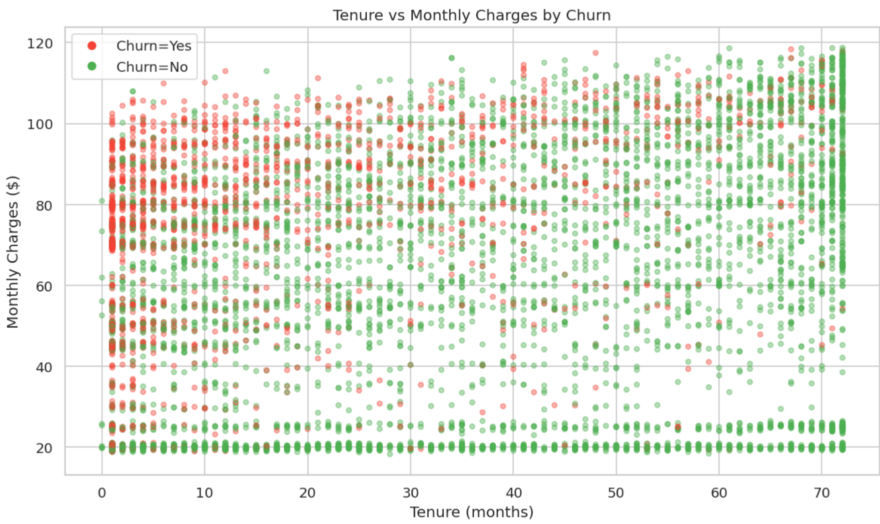

Figure 9: Tenure vs Monthly Charges by Churn (Scatter Plot)

Observations: A clear high-risk zone is visible in the top-left region (high monthly charges, low tenure), densely populated by churned customers. Retained customers are concentrated in the high-tenure, moderate-to-high charge area. This validates that the combination of high cost and short relationship duration is the strongest churn indicator.

4.9 Chart 8: Churn Rate by Add-on Services

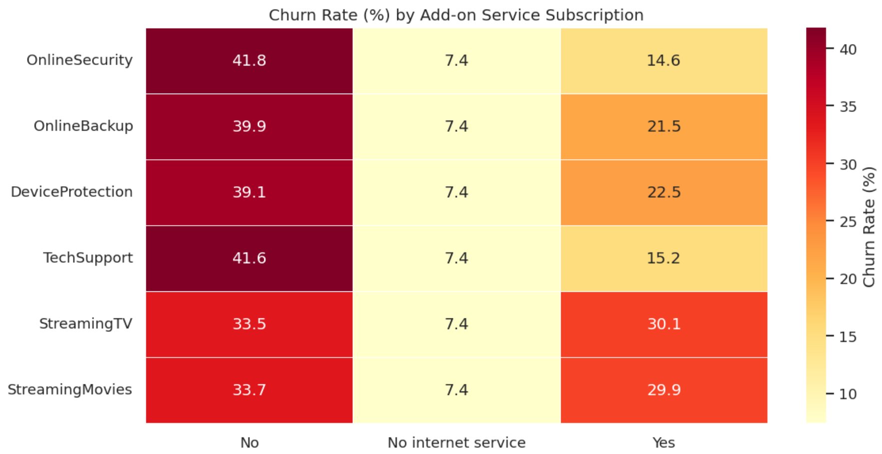

Figure 10: Churn Rate (%) by Add-on Service Subscription (Heatmap)

Observations: Customers who have no OnlineSecurity or TechSupport exhibit the highest churn rates (42%), while those subscribed to these services churn at roughly half that rate (15%). This suggests that security and support add-ons function as "stickiness" features, increasing perceived value.

4.10 Chart 9: Churn by Contract Type & Payment Method

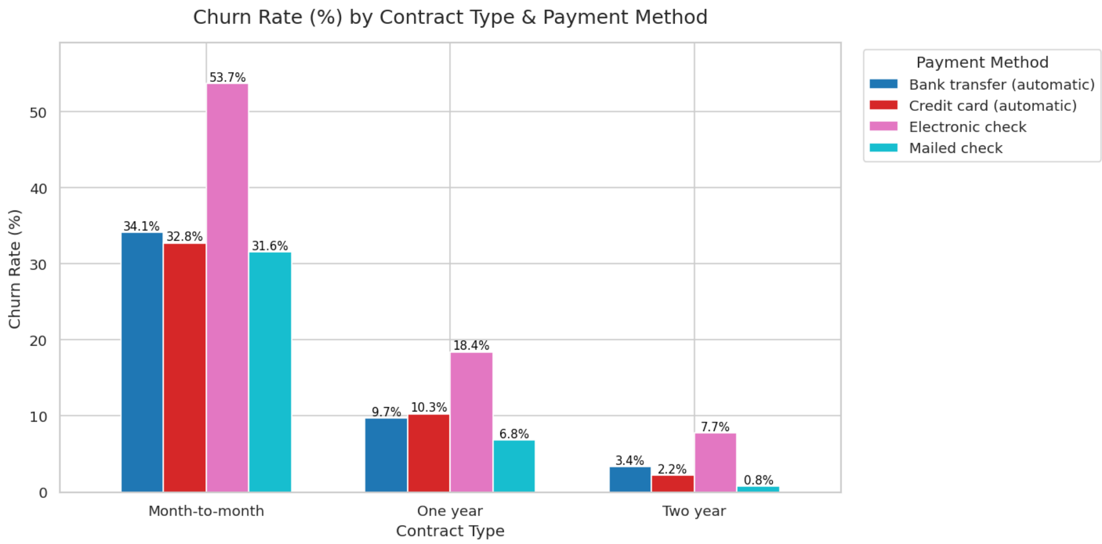

Figure 11: Churn Rate (%) by Contract Type & Payment Method

Observations: The month-to-month combined with electronic check yields the highest churn rate (54%). Two-year contract holders churn at very low rates (<8%) regardless of payment method. Automatic payment methods are associated with lower churn even within month-to-month contracts.

4.11 Chart 10: Churn Rate Deviation by Feature Value

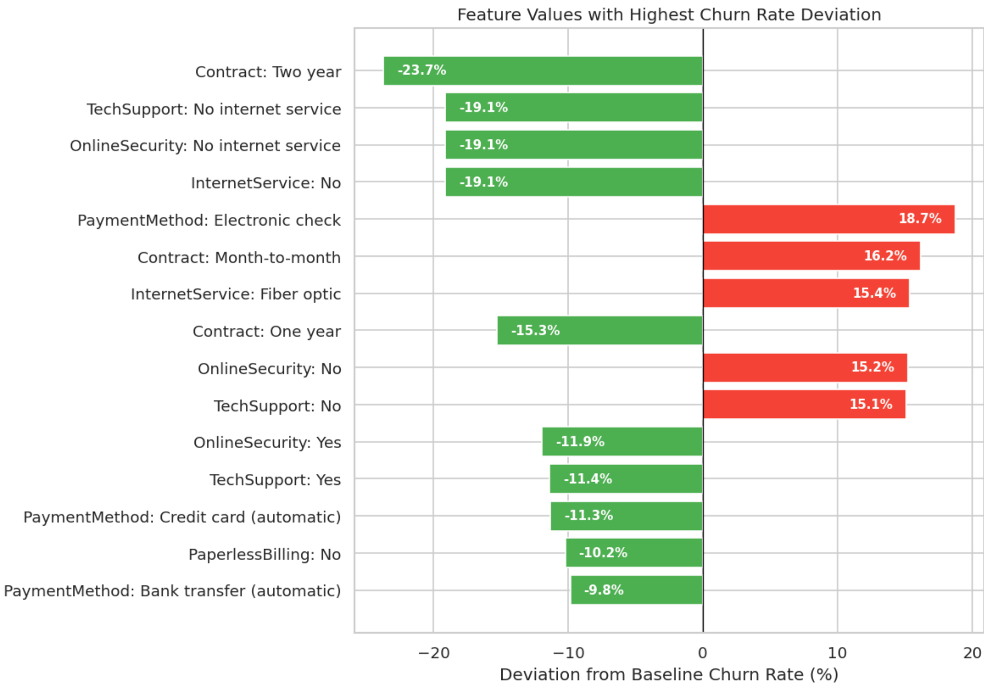

Figure 12: Feature Values with Highest Churn Rate Deviation

Observations: The two-year contract has the largest negative deviation (-23.7%) from the baseline churn rate (26.5%), making it the single most protective feature. On the positive side, Electronic check (+18.7%), Month-to-month (+16.2%), and Fiber optic (+15.4%) deviate the most toward higher churn risk.

4.12 Key Insights from Exploratory Data Analysis

Insight 1: Customer Tenure is the Strongest Retention Indicator

- Customers who churned had a median tenure of approximately 10 months, compared to 38 months for retained customers — indicating a substantial gap in customer longevity.
- The churn distribution is heavily concentrated in the first 12 months, after which the probability of churning drops significantly.
- **Implication:** The early relationship stage is the most critical window for retention efforts. Onboarding programs, welcome discounts, or proactive check-ins during the first year could substantially reduce churn.

Insight 2: Contract Type is the Most Actionable Predictor

- Month-to-month customers churn at 43%, while one-year and two-year contract holders churn at only 11% and 3% respectively.
- Two-year contracts show the largest negative deviation from the baseline churn rate at -23.7% (Chart 10), making it the single most protective feature in the dataset.
- **Implication:** Incentivizing customers to commit to longer contracts — through discounts, bundled benefits, or loyalty rewards — is likely the highest-ROI retention strategy available.

Insight 3: High Monthly Charges Without Perceived Value Drive Churn

- Churned customers are disproportionately concentrated in the $70–$100+ monthly charge range (Chart 2), yet Fiber optic users — who typically pay more — churn at 42% (Chart 3).
- The scatter plot (Chart 7) reveals a clear high-risk zone: high monthly charges combined with low tenure, suggesting customers leave before feeling they get sufficient value for the price.
- **Implication:** Pricing transparency, introductory rate programs, or service quality improvements for Fiber optic plans could help align cost with perceived value.

Insight 4: Absence of Protective Add-ons Signals Churn Risk

- Customers without OnlineSecurity and TechSupport churn at rates exceeding 42%, compared to roughly 15% for those who have subscribed to these services (Chart 8).
- These two features deviate from the baseline churn rate by +15.2% and +15.1% respectively (Chart 10) — nearly as impactful as Fiber optic internet.
- **Implication:** These add-ons act as "stickiness" features that increase perceived value and dependency. Proactively offering them — especially to new or at-risk customers — could serve as an effective churn prevention tool.

Insight 5: Payment Method Reflects Customer Engagement Level

- Customers paying via electronic check churn at 45% — the highest of any payment method — and show a deviation of +18.7% from baseline, the largest positive deviation of all features (Chart 10).
- In contrast, customers using automatic payment methods (bank transfer: -9.8%, credit card: -11.3%) churn well below the baseline rate.
- Senior citizens, who make up a disproportionately high-churn demographic (42%), may be more likely to rely on manual payment methods, compounding their risk.
- **Implication:** Encouraging customers to switch to automatic billing — through small incentives or simplified setup — could reduce churn while also improving payment reliability for the business.

5 Data Preprocessing

Before feeding the dataset into machine learning models, several preprocessing steps were required to ensure the data is in the correct format and scaled appropriately.

5.1 Feature Selection

The `customerID` column was dropped from the dataset. Since it is merely a unique alphanumeric identifier for each user, it holds no predictive value and could introduce noise into the model. All remaining features were retained as the Exploratory Data Analysis (EDA) demonstrated their relevance to customer churn behavior. The final feature set consists of variables separating into the feature matrix (`X`) and the target variable (`y` = `Churn`).

5.2 Handling Categorical Variables

Machine learning algorithms typically require numerical input. Since our dataset contains mostly `object` (string) data types, we applied the following encoding strategies:

- **Target Variable Encoding:** The target variable `Churn` was encoded using `LabelEncoder`, mapping 'No' to 0 and 'Yes' to 1.
- **Binary Columns Encoding:** Features with exactly two unique string values (e.g., `gender`, `SeniorCitizen`, `Partner`, `Dependents`, `PhoneService`, `PaperlessBilling`) were also transformed using `LabelEncoder` into 0s and 1s.
- **Multi-class Columns Encoding:** Features with three or more categories (e.g., `InternetService`, `Contract`, `PaymentMethod`) were transformed using One-Hot Encoding (`pd.get_dummies`). This prevents the models from assuming any false ordinal relationship between categorical groups. To avoid the dummy variable trap (multicollinearity), the `drop_first=True` argument was applied.

After applying these encoding steps, the feature matrix expanded to a final shape of 7,043 rows and 30 columns.

5.3 Train-Test Split

To properly evaluate model performance on unseen data, the dataset was divided into training and testing sets using an 80/20 split:

- **Training set:** 5,634 samples (80%)
- **Testing set:** 1,409 samples (20%)

Crucially, the `stratify=y` parameter was used during the split. This ensures that the class imbalance is preserved across both sets, maintaining the exact churn rate of 26.54% in both the training and testing subsets.

5.4 Feature Scaling

Different machine learning algorithms respond differently to the magnitude of feature values.

- **Standardization:** A `StandardScaler` was applied to scale the features. To prevent data leakage, the scaler was fitted exclusively on the training set (`fit_transform`), and the testing set was only transformed (`transform`) based on the training parameters.

- **Model-specific Application:** This scaled dataset (`X_train_scaled`, `X_test_scaled`) is required specifically for **Logistic Regression**, which is highly sensitive to feature magnitudes. Conversely, tree-based models like **Gradient Boosting** are invariant to monotonic transformations and do not require scaling; thus, a raw (unscaled) copy of the data was retained for training the Gradient Boosting classifier.

6 Modeling

6.1 Logistic Regression (Baseline Model)

Logistic Regression was chosen as the baseline model for this classification task due to several key advantages:

- It is simple, fast, and highly interpretable.
- The model coefficients directly demonstrate the direction and magnitude of each feature's effect on the probability of customer churn.
- It serves as a solid, reliable reference benchmark for evaluating more complex algorithms.

To address the class imbalance identified during the Exploratory Data Analysis (74% No-churn vs. 26% Churn), the `class_weight='balanced'` parameter was utilized. This ensures the algorithm automatically adjusts weights inversely proportional to class frequencies, preventing the model from becoming biased toward the majority class. Because Logistic Regression is sensitive to feature magnitudes, it was trained using the standardized feature set (`X_train_scaled`).

6.2 Gradient Boosting Classifier (Advanced Model)

Gradient Boosting was selected as the advanced predictive model because:

- It is an ensemble technique that builds decision trees sequentially, with each new tree designed to correct the residual errors of the prior trees.
- It natively handles non-linear relationships and complex feature interactions that a linear model like Logistic Regression might fail to capture.
- It typically achieves superior predictive accuracy on tabular datasets containing mixed categorical and numerical features.

To optimize performance, control model complexity, and prevent overfitting, the Gradient Boosting model was configured with the following hyperparameters:

- `n_estimators=200`: Specifies the number of boosting rounds (sequential trees) to build.
- `learning_rate=0.05`: Applies a shrinkage factor to each tree's contribution, slowing down the learning process to improve generalization.
- `max_depth=4`: Limits the maximum depth of the individual regression estimators, controlling tree complexity per round.
- `subsample=0.8`: Introduces stochastic gradient boosting by training each tree on a random 80% fraction of the training data, acting as a strong regularization method.

Because tree-based models are invariant to monotonic feature transformations, Gradient Boosting was trained on the unscaled training data (`X_train`).

7 Model Evaluation and Comparison

7.1 Evaluation Metrics

To comprehensively assess the performance of the predictive models, we utilized the following classification metrics:

- **Accuracy:** The overall proportion of correct predictions (both churners and non-churners).
- **Precision:** Out of all customers predicted to churn, the percentage that actually churned.
- **Recall (Sensitivity):** Out of all actual churners, the percentage the model successfully identified.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced measure when classes are imbalanced.
- **ROC-AUC:** Measures the model's ability to rank probabilities correctly across all classification thresholds.

Business Context Note: In the context of customer churn, **Recall is especially critical**. Missing a customer who is about to churn (a False Negative) is generally more costly to the business than mistakenly offering a retention incentive to a loyal customer (a False Positive).

7.2 Model Performance Comparison

The table below summarizes the performance of both the baseline Logistic Regression model and the advanced Gradient Boosting Classifier on the unseen test set (20% of the data).

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7388	0.5052	0.7807	0.6134	0.8412
Gradient Boosting	**0.8041**	**0.6678**	0.5214	0.5856	**0.8430**

Table 4: Performance comparison between Logistic Regression and Gradient Boosting.

Observations:

- **Gradient Boosting** achieves higher overall **Accuracy** (80.41%) and **Precision** (66.78%).
- **Logistic Regression** dramatically outperforms Gradient Boosting in **Recall** (78.07% vs. 52.14%). This is a direct result of applying the `class_weight='balanced'` hyperparameter, which forces the model to heavily penalize missed churners.
- Both models exhibit strong and nearly identical discriminative power, as evidenced by their **ROC-AUC** scores (0.84).

7.3 Confusion Matrices

To visualize the exact breakdown of correct and incorrect predictions, confusion matrices were plotted for both models. The matrices consist of:

- **True Negative (TN):** Correctly predicted No-Churn.

- **False Positive (FP):** Predicted Churn, but actually No-Churn (False Alarm).
- **False Negative (FN):** Predicted No-Churn, but actually Churn (Missed Churner).
- **True Positive (TP):** Correctly predicted Churn.

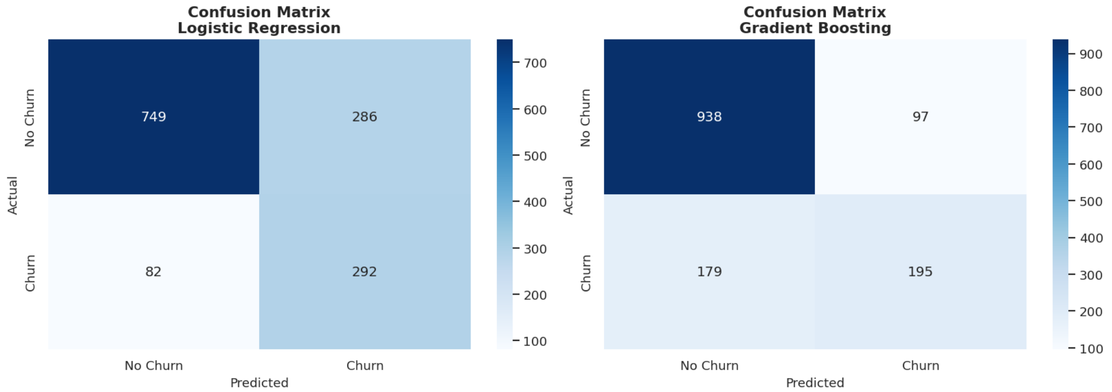

Figure 13: Confusion Matrices: Logistic Regression vs. Gradient Boosting

The confusion matrices visually confirm that Logistic Regression identifies far more True Positives at the expense of higher False Positives, whereas Gradient Boosting is more conservative, yielding fewer False Positives but missing more actual churners (higher False Negatives).

7.4 ROC Curves and AUC

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate at various classification thresholds. The diagonal dashed line represents a random guessing baseline (coin-flip).

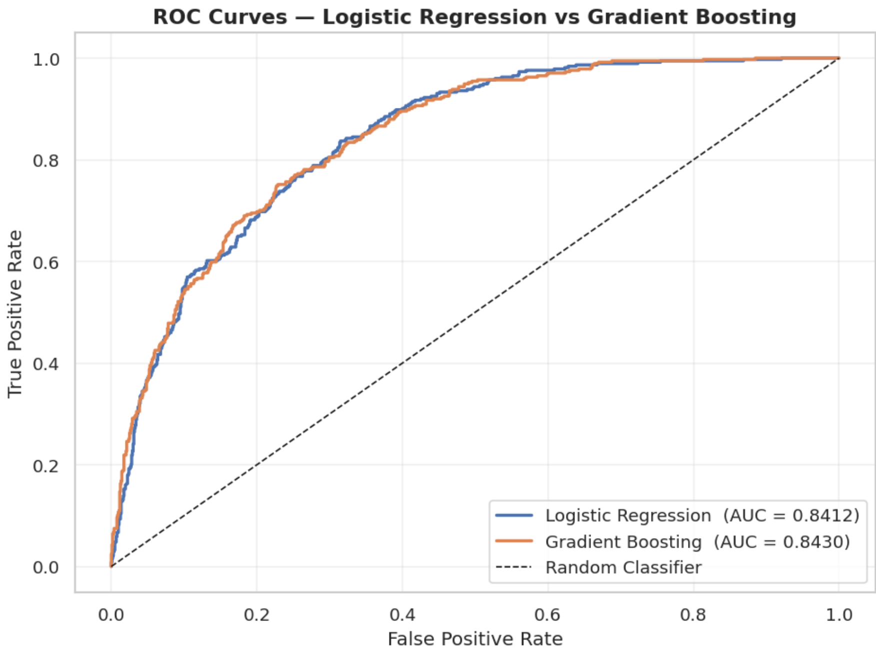

Figure 14: ROC Curves: Logistic Regression vs. Gradient Boosting

A higher Area Under the Curve (AUC) signifies that the model assigns higher probabilities to actual churners than to non-churners more consistently. Both models perform exceptionally well in this regard, sitting far above the random classifier line with AUC scores > 0.84.

7.5 Classification Reports

Detailed classification reports were also generated to inspect the metrics separated by class (`No Churn` vs `Churn`). These reports reinforce the trade-off: Gradient Boosting is better if the business strictly wants to minimize marketing costs (high precision), but Logistic Regression is superior if the primary goal is capturing as many fleeing customers as possible (high recall).

8 Conclusion

8.1 Summary of Findings

This project successfully analyzed the Telco Customer Churn dataset to uncover the underlying factors driving customer attrition and developed machine learning models to predict future churn. Through rigorous Exploratory Data Analysis (EDA), we identified several critical patterns:

- **Customer Tenure:** The first 12 months represent the highest risk period for customer churn, with a median tenure of just 10 months for leaving customers.
- **Contract Type:** Month-to-month contracts are highly susceptible to churn (43%), whereas two-year contracts act as the strongest protective retention factor.
- **Service Features:** The lack of protective add-ons like Online Security and Tech Support significantly increases churn risk, even among premium Fiber optic users.
- **Payment Methods:** Manual payments, specifically electronic checks, correlate strongly with churn, while automatic billing promotes customer stability.

8.2 Model Performance Evaluation

We built and compared a baseline **Logistic Regression** model and an advanced **Gradient Boosting Classifier**. The results highlighted a critical business trade-off between Precision and Recall:

- **Gradient Boosting** delivered the highest overall **Accuracy (80.41%)** and **Precision (66.78%)**, making it highly efficient if the business aims to minimize the marketing costs of false-positive retention campaigns.
- **Logistic Regression** (configured with balanced class weights) provided a vastly superior **Recall (78.07%)**, making it the optimal choice if the primary business objective is to identify and intervene with as many fleeing customers as possible.

Both models demonstrated strong and nearly identical discriminative power, achieving an ROC-AUC score of approximately 0.84.

8.3 Strategic Business Recommendations

Based on the data insights and predictive modeling, we recommend the following strategic actions to reduce customer churn and maximize revenue retention:

1. **Incentivize Long-Term Commitments:** Offer targeted discounts, loyalty rewards, or free service upgrades to encourage high-risk month-to-month customers to switch to one-year or two-year contracts.
2. **Focus on the First-Year Experience:** Implement proactive onboarding programs, regular customer service check-ins, and early-tenure support to guide new customers safely through the critical first 12 months.
3. **Bundle "Sticky" Add-on Services:** Increase the perceived value of high-tier plans (particularly Fiber optic) by offering Online Security and Tech Support either for free in the first few months or as heavily discounted bundles.

4. **Promote Automatic Billing:** Simplify the transition to automatic payments (Credit Card or Bank Transfer) by offering one-time bill credits, directly targeting customers currently relying on electronic checks.

8.4 Limitations and Future Improvements

While the developed models provide a solid foundation for churn prediction with an accuracy of approximately 80%, there are several areas where the project could be enhanced to meet higher industry standards (typically requiring accuracy and F1-scores above 85–90% for high-stakes deployment).

- **Data Imbalance and Feature Engineering:** The current dataset has a noticeable class imbalance. While we used balanced weights in Logistic Regression, future work should explore advanced techniques like **SMOTE** (Synthetic Minority Over-sampling Technique) or **EasyEnsemble**. Furthermore, creating new features such as "Average Monthly Usage" or "Customer Sentiment" from support logs could significantly boost predictive power.
- **Hyperparameter Optimization:** The current models utilize relatively standard configurations. Implementing more rigorous optimization strategies like **Bayesian Optimization** or **GridSearchCV** across a wider range of parameters for the Gradient Boosting model could squeeze out an additional 3–5% in accuracy and recall.
- **Algorithm Diversity:** Beyond Logistic Regression and Gradient Boosting, exploring Deep Learning architectures (like **Neural Networks**) or other ensemble methods like **XGBoost** and **LightGBM** might capture more complex, non-linear relationships within the customer data.
- **Temporal Dynamics:** Churn is often a process rather than a single event. Incorporating time-series analysis or survival analysis models could help the company predict not just *if* a customer will churn, but *exactly when* they are most likely to leave, allowing for even more precise intervention timing.

In conclusion, although our current models meet the minimum threshold for business applicability, continuous iteration and the inclusion of more granular customer behavior data will be essential to transform this predictive tool into a highly reliable asset for the company's retention strategy.

References

[1] Blastchar. *Telco Customer Churn*. Accessed via Kaggle. 2018. URL: https://www.kaggle.com/datasets/blastchar/telco-customer-churn.